

Unit 2 : Data Handling & Preprocessing Artificial Intelligence & Data Science

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What is Data Preprocessing?

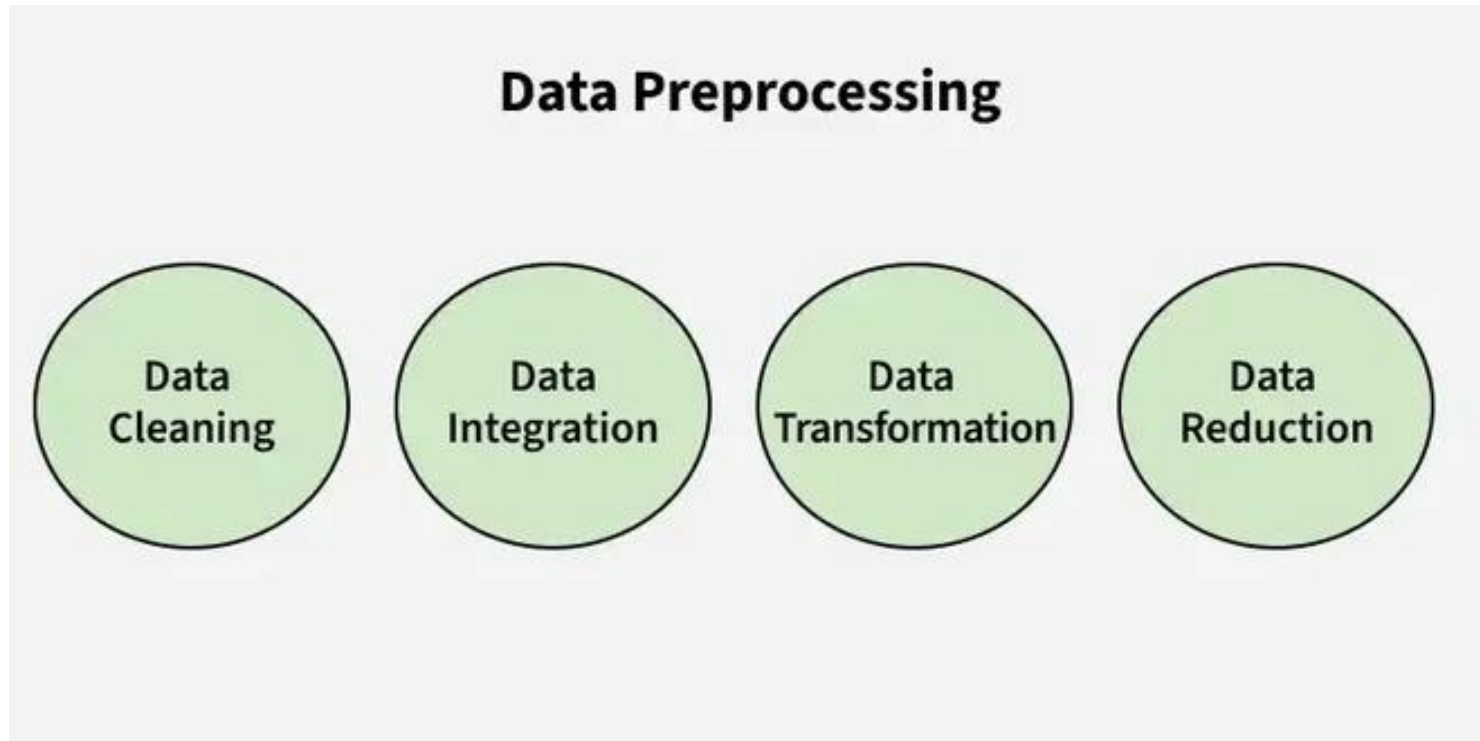
Process of preparing raw data for analysis and machine learning models.

- **Definition:** Transforming raw data into clean, usable format
- **Challenge:** Real-world data is messy, incomplete, inconsistent
- **Goal:** Prepare high-quality data for analysis
- **Impact:** Quality preprocessing = Better model performance

Importance of Data Preprocessing

- **Quality**-Removes errors and inconsistencies from raw data.
- **Efficiency** Reduces data size and improves processing speed.
- **Accuracy**-Improves model accuracy and prediction reliability

Complete Preprocessing Pipeline



Steps in Data Preprocessing

1. **Data Collection**
2. **Data Cleaning** (remove noise, fix errors, handle missing values)
3. **Data Integration** (combine multiple sources if needed)
4. **Data Transformation** (normalize/scale, encode, aggregate)
5. **Data Reduction** (remove irrelevant/duplicate data, dimensionality reduction)
6. **Data Splitting** (train-test/validation)

For your text mining research (CG/CG style):

1. Collect data
2. Clean text (lowercase, remove URLs, stopwords, special chars)
3. Tokenize
4. Encode / vectorize (TF-IDF, embeddings, or BERT tokenizer)
5. Split dataset

NumPy - Numerical Computing

Python library for efficient numerical operations on arrays.

- **Arrays:** N-dimensional arrays for storing large datasets
- **Operations:** Fast mathematical operations on entire arrays
- **Speed:** 100x faster than native Python lists
- **Foundation:** Base library for Pandas and machine learning

NumPy in Action

```
import numpy as np

arr = np.array([1, 2, 3, 4, 5])

mean = np.mean(arr)
std = np.std(arr)

result = arr * 2  # Element-wise multiply
matrix = arr.reshape(5, 1)  # Reshape into 5x1 matrix

print("Mean:", mean)
print("Std:", std)
print("Result:", result)
print("Matrix:\n", matrix)
```

Pandas - Data Manipulation

Purpose Python library for working with tabular data (like Excel spreadsheets)

- **DataFrames:** Table-like data structures with rows and columns
- **Operations:** Filter, sort, group, and merge data easily
- **File I/O:** Load data from CSV, Excel, SQL databases
- **Data Cleaning:** Handle missing values and duplicates

Step 1: Data Cleaning

- **Remove Duplicates:** Delete identical rows in dataset
- **Fix Data Types:** Correct mismatched data types
- **Standardize Format:** Consistent naming and formatting
- **Remove Outliers:** Identify and handle unrealistic values

Step 2: Missing Values - Detection

- **Identify:** Find NaN or null values in dataset
- **Visualize:** See patterns and extent of missing data
- **Analyze:** Understand why data is missing
- **Decide:** Drop or fill missing values appropriately

Handling Missing Values -Methods

Drop-Remove rows or columns

Fill-Replace with mean, median, mode, or forward fill

Impute-Use advanced methods like KNN or regression

Feature Engineering

Creating new features from existing data to improve model performance

- **Create:** Combine or transform existing features
- **Scale:** Normalize features to same scale
- **Encode:** Convert categorical variables to numbers
- **Select:** Keep only important features

Dimensionality Reduction

Reducing number of features while retaining important information

- **Why?** Remove redundant features, faster training
- **Benefits:** Reduces overfitting, improves model speed
- **Methods:** PCA, Feature Selection, Correlation Analysis
- **Trade-off:** Reduced features vs Information loss

Feature Selection Methods

Correlation

Remove highly correlated features

PCA

Principal Component Analysis

RFE

Recursive Feature Elimination

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